Zhiyuan Jia

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EDUCATION

University of Washington 09/2022 - 06/2026
Bachelor of Science, Computer Science and Applied Math (Double Major, GPA: 4.00/4.00) *Seattle, Washington*

- **Relevant Coursework:** Deep Learning, Machine Learning, Natural Language Processing, High-Performance Scientific Computing, Computer Vision, Advanced Linear Algebra, Finite Markov Chains, Probability Theory, Data Structures and Algorithms, Software Design and Implementation, Hardware-Software Interface.
- **Honors & Awards:** Dean's List (2022–2026) – 12 quarterly and 4 annual recognitions, summa cum laude.

University of Washington Expected 09/2026 - 06/2030
PhD Candidate, Industrial Engineering *Seattle, Washington*

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL, TypeScript, JavaScript, HTML, CSS, MATLAB
Technologies: PyTorch, Hugging Face, React, Node.js, LaTeX, Git, Gurobi

EXPERIENCE

University of Washington Department of Industrial & Systems Engineering 06/2025 - Present
Research Assistant, advised by Prof. Chaoyue Zhao and Prof. Yuxiang Peng *Seattle, Washington*

- Worked on semi-structured and unstructured sparsity in LLMs.
- Reformulated SparseGPT's masking and compensation as a mixed-integer programming problem.
- Restructured SparseGPT's pipeline to be row-wise and layer-wise, reducing peak GPU memory usage by 70%.
- Implemented a Quantum-Hamiltonian-Descent-based mask solver as an alternative to SparseGPT, reducing perplexity loss under semi-structured sparsity by over 30% and reconstruction error by 50%.

University of Washington Paul G. Allen School 12/2025 - 03/2026
Research Assistant, collaborating with Dr. Mars Gao and Dr. Yikun Zhang *Seattle, Washington*

- Worked on guided sampling strategies for LLMs to match reinforcement learning performance.
- Implemented block-wise power-sampling, reducing computation time by 35% and boosting single-shot accuracy on the MATH500 dataset by 10% compared to its original version.
- Proposed and implemented the usage of token-wise inverse conditional probability to determine suffix tokens for resampling, gaining an additional 10% reduction in computation time and 6% increase in accuracy.
- Proposed semantically block-wise sampling for better interpretability, and empirically verified its scale limitation.

University of Washington Department of Industrial & Systems Engineering 03/2024 - 05/2025
Research Assistant, advised by Prof. Chaoyue Zhao and Prof. Lei Fan *Seattle, Washington*

- Worked on Quantum Hamiltonian Descent's performance over non-smooth, non-convex optimization problems.
- Implemented QHD-driven Sequential Quadratic Programming algorithm, numerically verifying QHD's theoretical advantage in achieving global optimality.
- Implemented and automated experiments for 50+ benchmark functions and 2000+ test instances.
- Resulted in a paper submission to *Operations Research*, the flagship journal in the operations research field.

TECHNICAL PROJECTS

AI Art Detection: CNN vs Transformer | Python, PyTorch, OpenCLIP, timm

- Designed and implemented a binary image classification system to distinguish AI-generated and human-created artwork using ResNet-50 and ViT-B/16 backbones.
- Conducted a structured experiment across 12 architecture–pretraining–adaptation combinations.
- Built a full transfer learning pipeline with data preprocessing, augmentation, training, early stopping, and evaluation.
- Achieved 15% higher test accuracy than standard supervised baselines using ViT-B/16 + CLIP + linear probing.

PUBLICATIONS

- Jiaqi Leng, Yufan Zheng, **Zhiyuan Jia**, Lei Fan, Chaoyue Zhao, et al. *Quantum Hamiltonian Descent for Non-smooth Optimization*. *arXiv preprint* arXiv:2503.15878, 2025. (Submitted to *Operations Research*).